Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Page 1 of **??**

Exam Title Faculty Name Month YYYY

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STAMFORD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Term X Examination, Spr./Sum//Fall YYYY Semester
Course Code: Course Title
CT: Faculty Name
Date and Time: dd/MM/YYYY, HH:MM - HH:MM

Batch: CSE-S-XX-X **Campus:** Siddeswari
Duration: 2 hours **Full Marks:** 30 marks

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*(There are **FOUR** questions. **Answering all of them is mandatory**. Figures in the right margin indicate CLOs and marks. Writing anything on the question paper is strictly prohibited.)*

[noitemsep,topsep=0pt][1-1]Now, take a look at a figure. [1-1] And now look at the

□

Gambar 1: A basic OP-Amp circuit

two sub-figures. As a potential data scientist, try to find out the following:

.5 □

Gambar 2: A boxplot

.5 □

Gambar 3: A histogram

Gambar 4: EDA from the titanic dataset

1. (a) i. Sub task 1. (3)

   ii. Sub task 2.

   [noitemsep,topsep=0pt]

   (b) [1-1] Consider the following equation: (3)

   $$\text{net}_j = \sum_i W_{ij} x_{ij} = W_{0j} x_{0j} + W_{1j} x_{1j} + \ldots + W_{Ij} x_{Ij}$$

   Do the following tasks:

   i. Sub task 1.

   ii. Sub task 2.

   [noitemsep,topsep=0pt]

   (c) Task description (4)

[noitemsep,topsep=0pt][2-3]: The weights of the neural network are randomly set as

□

Gambar 5: Architecture of the neural network

Tabel 1: Weights of the neural network

$x_0 = 1$	$W_{0e} = 0.5$	$W_{0f} = 0.5$	$W_{0g} = 0.5$		$W_{0h} = 0.25$
$x_a = 0.8$	$W_{ae} = 0.5$	$W_{be} = 0.9$	$W_{ce} = 0.6$	$W_{de} = 0.6$	$W_{eh} = 0.9$
$x_b = 0.3$	$W_{af} = 0.5$	$W_{bf} = 0.9$	$W_{cf} = 0.6$	$W_{df} = 0.8$	$W_{fh} = 0.9$
$x_c = 0.6$	$W_{ag} = 0.4$	$W_{bg} = 0.8$	$W_{cg} = 0.6$	$W_{dg} = 0.6$	$W_{gh} = 0.8$
$x_d = 0.6$					

follows: The weights of the neural network are randomly set as follows:

2. (a) Now perform the following task (6)

   i. Calculate the value of $(1 + \frac{1}{\exp^{-x}})$.

   ii. Task 2.

   [noitemsep,topsep=0pt]

   (b) Take a look at the given Figure 1 and do task 1. (4)

[noitemsep,topsep=0pt][4-4]
(10)

3. ====== Best of Luck ======

Appendix

Tabel 2: T-table of Confidence interval estimate

t PERCENTAGE POINTS

ν	60.0%	66.7%	75.0%	80.0%	87.5%	90.0%	95.0%	97.5%	99.0%	99.5%	99.9%
1	0.325	0.577	1.000	1.376	2.414	3.078	6.314	12.706	31.821	63.657	318.31
2	0.289	0.500	0.816	1.061	1.604	1.886	2.920	4.303	6.965	9.925	22.327
3	0.277	0.476	0.765	0.978	1.423	1.638	2.353	3.182	4.541	5.841	10.215
4	0.271	0.464	0.741	0.941	1.344	1.533	2.132	2.776	3.747	4.604	7.173
5	0.267	0.457	0.727	0.920	1.301	1.476	2.015	2.571	3.365	4.032	5.893
6	0.265	0.453	0.718	0.906	1.273	1.440	1.943	2.447	3.143	3.707	5.208
7	0.263	0.449	0.711	0.896	1.254	1.415	1.895	2.365	2.998	3.499	4.785
8	0.262	0.447	0.706	0.889	1.240	1.397	1.860	2.306	2.896	3.355	4.501
9	0.261	0.445	0.703	0.883	1.230	1.383	1.833	2.262	2.821	3.250	4.297
10	0.260	0.444	0.700	0.879	1.221	1.372	1.812	2.228	2.764	3.169	4.144
11	0.260	0.443	0.697	0.876	1.214	1.363	1.796	2.201	2.718	3.106	4.025
12	0.259	0.442	0.695	0.873	1.209	1.356	1.782	2.179	2.681	3.055	3.930
13	0.259	0.441	0.694	0.870	1.204	1.350	1.771	2.160	2.650	3.012	3.852
14	0.258	0.440	0.692	0.868	1.200	1.345	1.761	2.145	2.624	2.977	3.787
15	0.258	0.439	0.691	0.866	1.197	1.341	1.753	2.131	2.602	2.947	3.733
16	0.258	0.439	0.690	0.865	1.194	1.337	1.746	2.120	2.583	2.921	3.686
17	0.257	0.438	0.689	0.863	1.191	1.333	1.740	2.110	2.567	2.898	3.646
18	0.257	0.438	0.688	0.862	1.189	1.330	1.734	2.101	2.552	2.878	3.610
19	0.257	0.438	0.688	0.861	1.187	1.328	1.729	2.093	2.539	2.861	3.579
20	0.257	0.437	0.687	0.860	1.185	1.325	1.725	2.086	2.528	2.845	3.552
21	0.257	0.437	0.686	0.859	1.183	1.323	1.721	2.080	2.518	2.831	3.527
22	0.256	0.437	0.686	0.858	1.182	1.321	1.717	2.074	2.508	2.819	3.505
23	0.256	0.436	0.685	0.858	1.180	1.319	1.714	2.069	2.500	2.807	3.485
24	0.256	0.436	0.685	0.857	1.179	1.318	1.711	2.064	2.492	2.797	3.467
25	0.256	0.436	0.684	0.856	1.178	1.316	1.708	2.060	2.485	2.787	3.450
26	0.256	0.436	0.684	0.856	1.177	1.315	1.706	2.056	2.479	2.779	3.435
27	0.256	0.435	0.684	0.855	1.176	1.314	1.703	2.052	2.473	2.771	3.421
28	0.256	0.435	0.683	0.855	1.175	1.313	1.701	2.048	2.467	2.763	3.408
29	0.256	0.435	0.683	0.854	1.174	1.311	1.699	2.045	2.462	2.756	3.396
30	0.256	0.435	0.683	0.854	1.173	1.310	1.697	2.042	2.457	2.750	3.385
35	0.255	0.434	0.682	0.852	1.170	1.306	1.690	2.030	2.438	2.724	3.340
40	0.255	0.434	0.681	0.851	1.167	1.303	1.684	2.021	2.423	2.704	3.307
45	0.255	0.434	0.680	0.850	1.165	1.301	1.679	2.014	2.412	2.690	3.281
50	0.255	0.433	0.679	0.849	1.164	1.299	1.676	2.009	2.403	2.678	3.261
55	0.255	0.433	0.679	0.848	1.163	1.297	1.673	2.004	2.396	2.668	3.245
60	0.254	0.433	0.679	0.848	1.162	1.296	1.671	2.000	2.390	2.660	3.232
∞	0.253	0.431	0.674	0.842	1.150	1.282	1.645	1.960	2.326	2.576	3.090
ν	60.0%	66.7%	75.0%	80.0%	87.5%	90.0%	95.0%	97.5%	99.0%	99.5%	99.9%